Control Concept for L20A Output During Island Operation

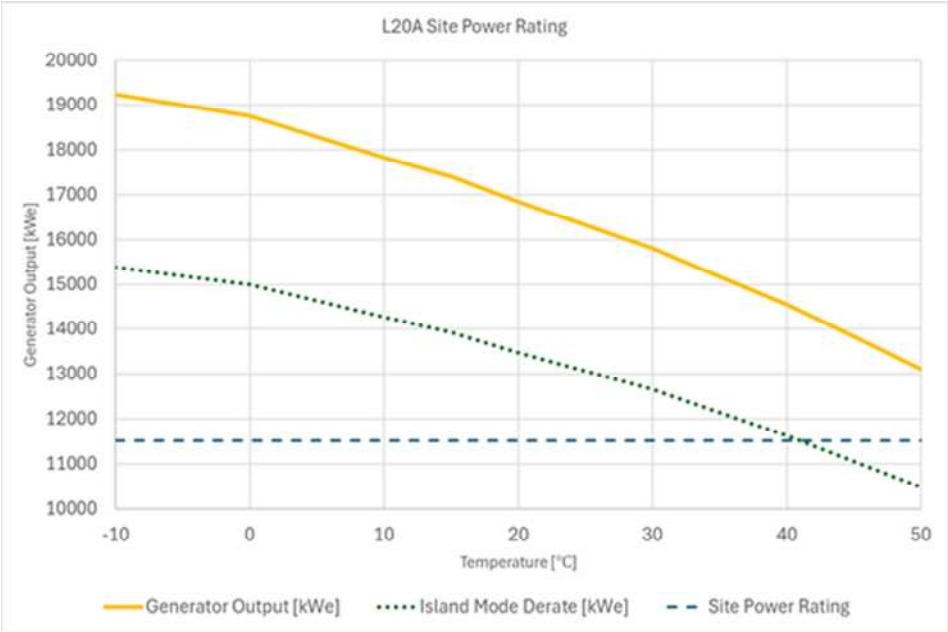
1. Purpose

This document summarizes the concept of output control and protection during island operation (non-grid-connected operation), as well as the requirements for S&S.

2. Requests from S&S

If output is limited throughout the year based on rated output at the annual maximum ambient temperature plus margin during standalone operation (blue dotted line below), the upper limit of output is significantly reduced.

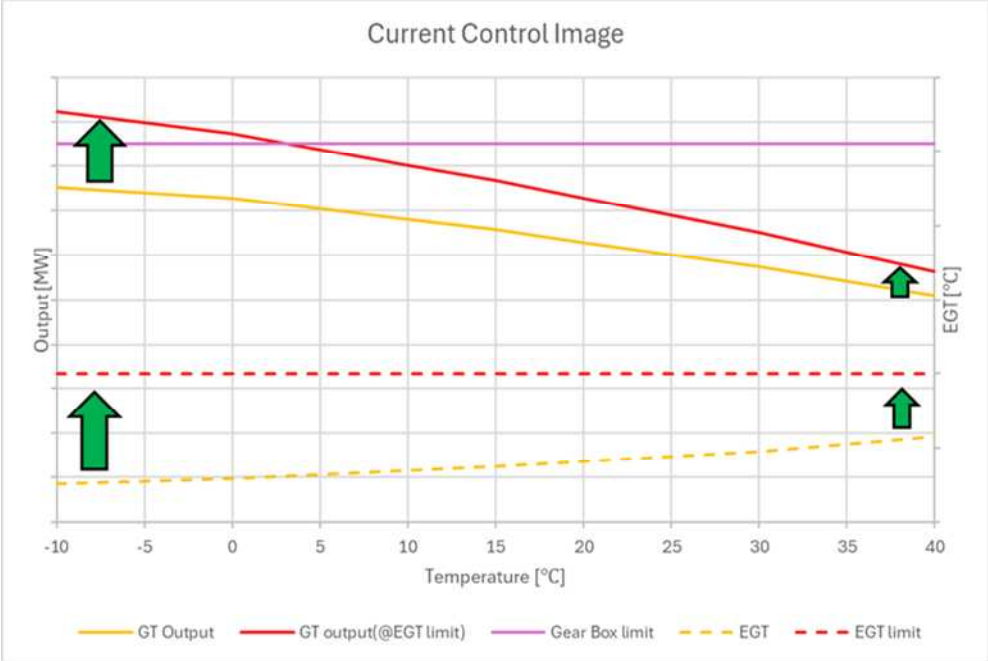
S&S believes there may be alternative technical or rating approaches that could better reflect real operating conditions while still maintaining appropriate design margins.



3. KHI Internal Study Results

3-1. Current Understanding

- **During grid-connected operation:**
  GT output is controlled not to exceed the rated output curve by GT control system, and the risk of overload is low.

- **During island operation:**
  GT output depends directly on the load and cannot be actively controlled by GT control system. Consequently, it is possible that the load increases without limitation depending on demand, leading to the risk of overload and potential damage to the engine, generator, and reduction gearbox.
- Under the current control scheme, EGT trip protection is provided and is activated when EGT reaches a fixed limit value that is independent of intake air temperature. However, for a rated TIT, EGT tends to decrease as intake air temperature decreases. In particular, at low intake temperatures, the margin between the actual EGT and the EGT trip limit becomes very large (See below fig.) Consequently, GT could be subject to significantly higher overload conditions before an EGT trip occurs, increasing the risk of gas turbine (GT) damage. Thus, this raises a concern about insufficient practical GT protection.
- In addition, since GT itself is capable of producing higher output at lower intake temperatures, there is also a concern that, if unexpectedly high load is applied under such conditions, GT may continue operating at excessively high output levels without triggering an EGT trip. Although EGT remains within the permissible range, the resulting high torque may cause damage to mechanical components such as reduction gearbox gears. This is one example of potential risks associated with the current control and protection philosophy when load flexibility is allowed in accordance with seasonal demand variations.
- 

- (NOTE)This figure is for reference only. Actual line will depend on site conditions.

3-2. Basic Policy

1. **Load shedding by S&S (or the end user) is a prerequisite during island operation.**
    - A system must be implemented by S&S or the end user to detect disconnecting from the grid and automatically shed low-priority loads (selective load shedding system) — **mandatory requirement**.
2. **In addition, KHI (GT side) shall implement additional GT protection control**, assuming possible operational errors or delayed responses on the S&S or customer side.

3-3. Concept of Output and Load Control (Proposed)

Stage 1: Alarm (Warning)

- When the green dotted line (derated output during island operation) is exceeded:
    - Issue an alarm
    - Display warning message: *"Please reduce load."*

Stage 2: Load Shedding Command

- When the yellow line (rated output) is exceeded
  * Continuous operation in the region exceeding GT rated output is, in principle, not permitted during island operation.
    - KHI sends a load shedding command (breaker open signal) to S&S.
    - Upon receiving this signal, S&S shall perform load shedding — **mandatory**.

Stage 3: Final Trip (GT Protection)

- If the above measures are not implemented and overload continues:
    - Stop (trip) the GT.
- GT protection should take priority even if there is a risk of blackout.
- Trip criteria (based on output or EGT) will be examined and defined later.

3-4. Concept of Output and Control Lines (Proposed)

- A uniform limitation similar to stand-by gas turbine generator (blue dotted line) shall not be used.

- The green dotted line, dependent on intake air temperature, shall be adopted as the steady-state target output line.
  This is intended to prevent load exceeded from GT rated output (yellow line) even during transient load increases (e.g., motor starts).
- Numerical values (percentage of rated output, margin to trip, time delay) will be examined and defined later.

3-5. Implementation and Operational Policy

- In case of blackout (including GT trip during island operation), power demand is assumed to be covered by UPS or stand by generators (S&S or the end user's scope).
- A policy prioritizing GT protections proposed to S&S in order to reduce risks of GT damage which requires long downtime for restoration.
- Final site conditions (altitude, atmospheric pressure, ambient temperature, fuel) must be provided (responsibility of S&S or end user), as they are essential for output and derating settings.
- Rated lines, derating lines, and protection lines shall be configured prior to shipment of the PLC from KHI.
- Parameter changes will be required during both S&S factory tests and on-site (end-user site) tests.
- Parameter changes shall, in principle, be performed by KHI under a separate contract.

3-6. Summary of Key Points

- During island operation, load shedding (selective load tripping) by S&S or the end user is a prerequisite.
- The intake-temperature-dependent green dotted line shall be used as the target output line during island operation.
- Proposal to prioritize GT protection.
- Three-stage GT protection shall be implemented: alarm, load shedding, and trip.

END